TfL API Endpoint Usage Overview

Purpose

This document outlines the Transport for London (TfL) Unified API endpoints used in the project and explains how each endpoint contributes to building and operating the live and hypothetical Underground network simulation.

1. Line Information Endpoint

Endpoint: /Line/Mode/tube

This endpoint provides a list of London Underground lines, including their identifiers and names. In our project, it is used by fetchAllLines.js to retrieve all Tube lines, which are then passed into buildFromRoutes.js. For each line, the project fetches its route sequence to extract stations and create station-to-station connections. Each connection is labelled with the relevant TfL line ID in the Neo4j *CONNECTS_TO* relationship.

2. Route Sequence Endpoint

Endpoint: /Line/{lineId}/Route/Sequence/all

This endpoint returns the ordered route sequence for a specific Underground line, including the stop points served by that line. In our project, it is called for each line ID returned by /line/Mode/tube. The station data from each route sequence is used to create Station nodes in the Neo4j database. Consecutive stations in the sequence are then used to create *CONNECTS_TO* relationships between stations. The project also calculates the straight-line distance between each pair of connected stations using their latitude and longitude coordinates and stores that distance on the edge.

3. Line Status Endpoint

Endpoint: /Line/Mode/tube/Status

This endpoint returns the current service status for all London Underground lines, including whether each line has a good service, delays, closures or other disruption information. In our project, it is used to display live line delay information in the frontend sidebar. This helps users see the current real-world operating status of each Underground line while using the simulator. In the What-If mode, this data is overridden by user-defined scenarios.

4. Detailed Line Status Endpoint

Endpoint: /Line/Mode/tube/Status?detail=true

This endpoint returns the current service status for each London Underground line, with additional disruption details included when a line is affected. When available, the disruption data can include affected routes, directions and stop points within the affected route section. In our project, this endpoint is used to identify live line conditions such as delays, closures and partial service disruption. This information helps the frontend determine which lines should be treated as closed or partially affected when the simulator is using real TfL network conditions.

5. Station Disruption Endpoint

Endpoint: /StopPoint/Mode/tube/Disruption

This endpoint returns live disruption information for Underground stations. In our project, it is used to detect live station-level disruptions or closures, such as full station closures, partial closures or access issues. This data can then be used to mark affected stations on the map and prevent closed stations being used in route planning.

6. Train Arrivals Endpoint

Endpoint: /Line/{lineIds}/Arrivals

This endpoint returns live arrival predictions for one or more London Underground lines. In our project, it is used to retrieve live train arrival data, including line IDs, destination information, expected arrival times, current locations and vehicle identifiers when available. The project uses this data to estimate and display live train positions on the map. If live arrival data is unavailable or does not contain usable train identifiers, the system falls back to simulated train movement based on the stored network graph.

Summary

Together, these endpoints define the Underground network’s structure (lines, stations, connections) and its operational state (live or simulated). This integration enables real-time routing as well as controlled “what-if” disruption analysis without modifying or reproducing any proprietary TfL assets.